

Mid Semester Exam

Subject: Multicore Architecture and Programming

Code: CP4292

Section A

1. What are the key differences between single core and multi-core architectures?
2. Define SIMD and MIMD systems in the context of multi-core processors.
3. List the types of communication mechanisms available for threads.
4. Identify the methods for handling data and functional parallelism in OpenMP.
5. Define the key constructs used in MPI programming.

Section B

1. Explain the transition from single core to multi-core architectures and its implications.
2. Compare SIMD and MIMD systems in terms of their architecture and applications.
3. Explain the role of message queues in thread communication.

Section C

1. Analyse the impact of interconnection networks on the performance of multi-core processors.
2. Evaluate the challenges faced in parallel program design for multi-core architectures.